



**IILM UNIVERSITY**

School of Computer Science and Engineering, Greater Noida, U.P.

## INTERNSHIP TOPIC

# Development, Deployment and Maintenance of the ThumbbyX Web Application

*A Full-Stack MERN Application, Built and Deployed on AWS*

**STUDENT NAME**

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**ENROLLMENT NUMBER**

**CS-2341847**

**PROGRAM & BATCH**

**B.Tech CSE, 2023–2027**

**ORGANIZATION**

**ThumbbyX Global Pvt. Ltd.**

# Introduction

*Practical exposure to the complete software development lifecycle*

- Worked as a Software Developer Intern in the Technology & Product Department at ThumbbyX Global Private Limited.
- Primary responsibility: development, deployment, maintenance, debugging and enhancement of the ThumbbyX web application.
- Built the ThumbbyX website (thumbbyx.com) from the ground up using the MERN stack and deployed it on AWS.
- Work continued after launch — maintaining the software, resolving issues, and adding features as requirements evolved.
- Bridged classroom knowledge from the B.Tech CSE program with real-world, production-grade engineering practice.

# 3

**Months**

Internship duration (Jun – Sep 2026)

# MERN

**Stack**

React, Node.js, Express.js, MongoDB

# AWS

**Cloud**

Production deployment platform

# Organization Profile

*ThumbbyX Global Private Limited*

ThumbbyX is a technology-driven startup working in construction technology, building a digital platform that connects construction professionals — contractors, civil engineers, and related trades — with verified profiles and relevant project opportunities.

Incorporated in 2025 and registered in Bihar, the company is classified under computer programming and consultancy activities. During the internship, I worked in the Technology & Product Department building and improving the core web platform, from initial development through to production deployment and ongoing maintenance.



## Website

thumbbyx.com



## Email

info@thumbbyx.com



## Registered

Private limited company, Bihar (est. 2025)

## BETA LAUNCH CITIES

- Delhi NCR
- Noida
- Greater Noida
- Gurugram
- Patna

*Platform enables professionals to build digital profiles, undergo verification, and access relevant project opportunities.*

# Problem Statement

*The existing ThumbbyX build was incomplete and not production-ready*

1

## Unstable backend

Backend requests could repeatedly crash the server.

2

## Incomplete frontend

Several UI parts still needed to be built.

3

## Weak integration

Frontend and backend were not reliably connected.

4

## No deployment

Application had not been prepared for the cloud.

5

## Reliability gaps

Needed testing and debugging before/after launch.

6

## Future-readiness

Codebase had to support upcoming feature growth.

# Project Objectives

*What the internship aimed to achieve, end to end*

- ✓ Develop the ThumbbyX web application from scratch
- ✓ Build the frontend using React.js with full user-facing functionality
- ✓ Develop backend services and APIs using Node.js and Express.js
- ✓ Integrate MongoDB for persistent data storage
- ✓ Unify frontend, backend and database into one full-stack app
- ✓ Deploy the completed application on Amazon Web Services (AWS)
- ✓ Test the application and identify issues during/after deployment
- ✓ Debug and resolve software problems as they occurred
- ✓ Maintain the deployed application and its existing functionality
- ✓ Implement feature upgrades as project requirements changed

# Scope of the Project

*From ground-up development through production deployment and upkeep*

**FE**

## Frontend Development

React.js pages, reusable components, forms, navigation and interactive elements.

**BE**

## Backend Development

Node.js + Express.js APIs handling requests, logic and responses to the frontend.

**DB**

## Database Integration

MongoDB used to create, retrieve, update and manage application data.

**AWS**

## AWS Deployment

Production environment configured, application deployed and verified live.

**MT**

## Maintenance & Upgrades

Ongoing debugging, testing and feature enhancement post-launch.

# Technologies & Tools Used

*The MERN stack, supported by AWS for cloud deployment*



## MongoDB

Database — persistent storage & management of app data



## Express.js

Backend framework — API dev, routing, request handling



## React.js

Frontend library — interactive, component-based UI



## Node.js

Runtime — executes backend JavaScript services



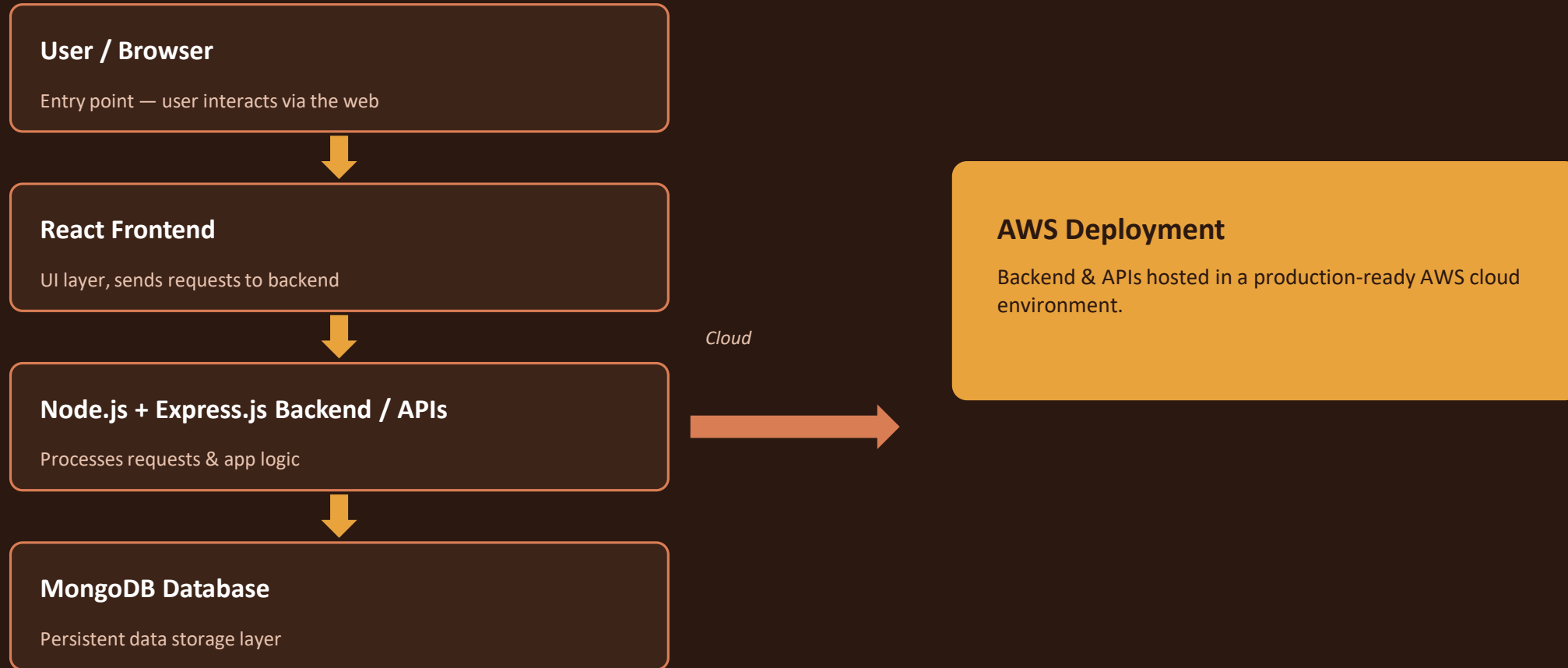
## AWS

Cloud platform — deployment & hosting of the application

*Full-stack architecture: React, Node.js, Express.js and MongoDB were integrated together, then deployed and hosted on AWS.*

# System Architecture

*Client-server, full-stack MERN architecture deployed via AWS*



# Methodology

*An iterative development approach across ten repeated stages*



# Major Internship Responsibilities

*Key areas of ownership across the internship*

WD

## Web Development

Built the ThumbbyX website from scratch using the MERN stack.

FE

## Frontend

Implemented user-facing interfaces & interactivity with React.js.

BE

## Backend

Developed server-side functionality and APIs with Node.js & Express.js.

DB

## Database

Integrated MongoDB for persistent application data.

AWS

## Deployment

Deployed the application on AWS.

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## Debugging

Investigated and fixed software issues and errors.

MT

## Maintenance

Maintained deployed software and verified functionality.

UP

## Feature Upgrades

Added and modified features per evolving requirements.

# Expected Outcomes

*What the internship delivered for ThumbbyX and for my own growth*

- ✓ A fully functional ThumbbyX web application, developed from scratch
- ✓ Successful integration of frontend, backend and database via MERN
- ✓ Successful deployment of the application on AWS
- ✓ A maintainable system that can be debugged and modified over time
- ✓ Continuous improvement through ongoing feature upgrades
- ✓ Practical experience in full-stack development and cloud deployment
- ✓ Stronger understanding of debugging, testing and production practices

# Thank You

*Development, Deployment and Maintenance of ThumbbyX — MERN Stack & AWS*

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Internship Guide: Ms. Surabhi Purwar, School of Computer Science and Engineering